

# Positional Dominance in Network Games: Equilibrium Conditions for Strategic Abstention from Velocity Competition

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## VERSION NOTE — WHAT CHANGED IN V2, AND WHY

Version 1 contained a defect in the derivation of the analytical threshold. It is corrected here. The numerical results are unaffected; the analytical result is superseded; one attribution is withdrawn.

**1. The value gap was truncated inconsistently.** Section 3 of v1 stated  $\Delta V(\sigma) \approx a \cdot \sigma^2 - b$ , a quadratic with no linear term. Appendix A of v1 derived  $\Delta V(\sigma) = (\beta_J - \beta_V) \cdot \sigma \cdot E|I_A - I_B| - c(v^*)$ , a linear function with no quadratic term. These are different functions and cannot both be the derivation. Each is a partial truncation of the paper's own value-function ansatz.

**2. The intercept was justified incorrectly, and is zero as stated.** v1 argued " $b > 0$  because V earns positive rents at  $\sigma = 0$  from speed while J earns zero." At  $\sigma = 0$  the velocity player's optimum is  $v^* = 0$ , so V earns exactly zero, not positive rents; and if J likewise earns zero then  $b = 0$  and  $\sigma^* = \sqrt{b/a}$  collapses to zero. §3.1 restores the constant term  $\alpha_J - \alpha_V$  that the v1 derivation dropped, and identifies it with the capitalised fixed cost of holding position.

**3. The correction factor  $\psi \approx 0.50$  is withdrawn as an attribution.** With the linear term restored, the root of the full ansatz is *necessarily* below  $\sqrt{b/a}$ . Under the paper's own parameter formulas the omitted linear term is of the magnitude required to account for the entire gap between 0.665 and 0.33. The v1 apportionment — "approximately equally to storage optionality ( $\sim 25\%$ ) and Markov persistence ( $\sim 25\%$ )" — is therefore not identified and is retracted pending the test specified in §3.3.

**4. Existence is now proved without the approximation.** New §3.0 establishes existence of  $\sigma^*$  for the equilibrium value functions directly, via equilibrium existence in finite-state discounted stochastic games, continuity in  $\sigma$ , an exact sign at zero, and divergence — the last requiring  $\gamma > 0$ . §3.0.1 then proves **uniqueness** as well, via a Bellman convexity-preservation lemma together with the fact that the velocity player's payoff becomes exactly affine once the latency cap binds, under a checkable condition (C) on primitives. Neither result uses the ansatz. Uniqueness for the genuine two-state chain, as opposed to the comparative-static reading, remains open.

**5. Addendum Table A.1 status column is corrected.** Two rows previously labelled "Proved · 0 sorry · exact 1/3" are theorems of the form *rfl* on a definition, and a third holds for every closed interval. They are true,

machine-checked, and carry no information about the value  $1/3$ . The conjecture itself is unchanged: v1 stated it as open and unproved, and it remains so.

### Abstract

We study a two-player infinite-horizon stochastic game on a capacitated linear distribution network in which one player commits capital to control key nodes — pipelines and storage hubs — while the rival invests in low-latency execution infrastructure. Using Markov Perfect Equilibrium with supply–demand shocks governed by a two-state volatility Markov chain, we establish the existence of a threshold  $\sigma^*$  above which the positional player's equilibrium payoff strictly exceeds that of the velocity-optimizing player. The advantage arises from private inventory signals and flow-redirection optionality that reaction speed cannot replicate in high-volatility regimes. *In this version* existence of  $\sigma^*$  is proved directly for the equilibrium value functions, without any value-function approximation, under two conditions: that holding position carries a fixed cost, and that it confers a strictly positive private signal ( $\gamma > 0$ ). Uniqueness is obtained separately from the complete quadratic ansatz  $\Delta V(\sigma) = a \cdot \sigma^2 + m \cdot \sigma - b$ , whose positive root is unique by Descartes' rule and strictly below the value  $\sqrt{b/a} \approx 0.665$  reported in v1; the intercept  $b$  is identified as the capitalised fixed cost of holding position rather than as velocity rents at zero volatility. Value iteration on a calibrated  $8 \times 8 \times 8$  grid yields  $\sigma^* \in [0.30, 0.36]$ , unchanged. Under U.S. energy midstream calibration, historical Permian–Cushing volatility of  $0.35\text{--}0.42$  exceeds  $\sigma^*$  in approximately 68% of months over 2015–2024. Sensitivity analysis gives  $\sigma^* \in [0.24, 0.43]$  across  $\pm 20\%$  variation in two parameters, which we now read as evidence that  $\sigma^*$  is a property of a particular network under particular carrying costs rather than a universal constant.

## 1. Introduction

Competition for advantage in distribution and trading networks takes two broadly distinct forms. One party commits capital to control physical or informational chokepoints — storage, pipeline capacity, an internalised order book. The other invests in reacting faster to public information. The received view in the microstructure literature treats the second as the frontier of competition. This paper gives conditions under which it is not.

The Permian Basin–Cushing–Gulf Coast corridor provides the canonical example: hub operators consistently extract rents unavailable to participants without storage rights, regardless of execution speed. We formalise the conditions under which strategic abstention from velocity competition is not a second-best outcome but a payoff-dominant equilibrium strategy, and derive a computable volatility threshold  $\sigma^*$  above which the positional player strictly outperforms the velocity-investing player in Markov Perfect Equilibrium.

### 1.1 Related literature

**Supply chain network games.** Nagurney (2012, 2014) establishes Nash equilibria in supply chain networks via variational inequalities. These models assume all players compete over flows and do not consider asymmetric infrastructure types — positional control versus velocity investment.

**Speed competition and HFT.** Budish, Cramton and Shim (2015) model the HFT arms race as a prisoner's dilemma; Foucault, Hombert and Rosu (2016) analyse fast-trader information advantages. Neither framework accommodates a player who opts out of speed investment as a dominant strategy.

**Real options.** Dixit and Pindyck (1994) establish the value of waiting under uncertainty. This is the closest single-agent antecedent; the contribution here is the two-player equilibrium condition.

## 2. Model

Discrete time, infinite horizon, discount factor  $\delta$ . Three nodes  $A \rightarrow C \rightarrow B$ : production, hub, export. Player J owns hub C, chooses capacity utilisation  $K \in [0, \bar{K}]$  and flow  $q \in [-K, K]$ , observes a private inventory signal of strength  $\gamma$ , pays transport cost  $\tau$  per unit moved, **and pays a fixed carrying cost  $c_K > 0$  per period for holding capacity, incurred whether or not the hub is used.** Player V invests in speed  $v \geq 0$  at cost  $c(v) = v^2/2$ , observes public signals only, and captures a fraction  $(1-\gamma)$  of informational content subject to a saturation cap  $\min(v, 1)$ . Volatility  $\sigma_t$  follows a two-state Markov chain on  $\{\sigma_L, \sigma_H\}$  with matrix P.

Change from v1:  $c_K$  was implicit in v1's  $R_j|_{\sigma=0}$  term but never named or given a symbol. Naming it is what makes §3.1 possible.

| SYMBOL                | VALUE       | MEANING                         | SOURCE         |
|-----------------------|-------------|---------------------------------|----------------|
| $\beta$               | 8.5 \$/bbl  | Price impact sensitivity        | Kilian (2009)  |
| $\tau$                | 2.0 \$/bbl  | Transport cost                  | FERC Form 6    |
| $\bar{K}$             | 1.5 MMbbl/d | Max hub capacity                | EIA STEO 2024  |
| $\delta$              | 0.985       | Monthly discount factor         | 18% annual     |
| $\gamma$              | 0.55        | Private signal strength         | Pirrong (2012) |
| $\sigma_L / \sigma_H$ | 0.15 / 0.45 | Volatility states               | EIA monthly    |
| $P_{HH}$              | 0.75        | High-vol persistence            | WTI empirical  |
| $c_K$                 | calibrated  | Fixed carrying cost of capacity | see §3.1       |

## 3. The value gap

### 3.0 Existence of the threshold, without the ansatz

Proposition 1' below establishes existence and uniqueness *within* the quadratic value-function approximation. That is weaker than the paper needs, because the ansatz is assumed rather than derived. This section proves existence for the equilibrium value functions themselves. It uses no approximation, and it isolates the one condition the result actually requires.

#### ASSUMPTION A1 (SATISFIED BY THE COMPUTATION OF §4)

The state space is a finite grid; action sets are compact; per-period payoffs are bounded and continuous in (state, action,  $\sigma$ ); the transition kernel is continuous in  $\sigma$ ; and  $\delta \in (0,1)$ .

#### LEMMA 3.1 · EQUILIBRIUM EXISTS

Under A1 the game admits a stationary Markov perfect equilibrium. This is the finite-state discounted stochastic game case; existence in stationary strategies is Fink (1964) and Takahashi (1964), with the standard treatment in Filar and Vrieze (1997). The value iteration of §4 computes on an  $8 \times 8 \times 8$  grid and therefore satisfies the hypotheses directly.

#### LEMMA 3.2 · CONTINUITY

The equilibrium values  $V_J^*(\sigma)$ ,  $V_V^*(\sigma)$ , and hence  $\Delta V(\sigma)$ , are continuous in  $\sigma$ . The Bellman operator is a  $\delta$ -contraction on the space of bounded functions; its data are continuous in  $\sigma$  by A1; the fixed point of a contraction depending continuously on a parameter is continuous in that parameter, and the maximised values are continuous by Berge's theorem.

### LEMMA 3.3 · THE SIGN AT ZERO IS EXACT

$$\Delta V(0) = -c_K/(1-\delta) < 0.$$

At  $\sigma = 0$  the dislocation  $|I_A - I_B|$  contributes nothing to either payoff.  $V$ 's problem reduces to  $\max_v -v^2/2$ , attained at  $v^* = 0$  with value 0.  $J$  earns nothing and pays  $c_K$ . No approximation is used.  $\square$

### LEMMA 3.4 · DIVERGENCE, AND THE CONDITION IT REQUIRES

As  $\sigma \rightarrow \infty$  the two per-period payoffs are both asymptotically linear:

$$\Pi_J(\sigma) \sim \beta \cdot \kappa \cdot \sigma - \tau \cdot K, \quad \Pi_V(\sigma) \sim \beta \cdot (1-\gamma) \cdot \kappa \cdot \sigma - \frac{1}{2}$$

where  $\kappa = E|\xi|$ . Hence

$$\Delta V(\sigma) \rightarrow +\infty \iff \gamma > 0$$

$J$ 's benefit coefficient is  $\beta \cdot (K/\bar{K}) = \beta$  at the optimum  $K = \bar{K}$ ;  $V$ 's is  $\beta(1-\gamma)$ , since  $V$  observes only the public share of the signal and the capture term saturates at  $\min(v, 1) = 1$  for  $\sigma$  large. The gap in slopes is  $\beta\kappa\gamma > 0$  precisely when  $\gamma > 0$ .

This is the same quantity Appendix A obtains as  $\beta_J - \beta_V = \beta \cdot \gamma \cdot (1 - \delta \cdot P_{LL})^{-1}$ .  $\square$

Note that this is not automatic. Both players' payoffs grow at the same order in  $\sigma$ . What separates them is not that  $J$  holds the hub but that holding the hub confers a private signal. If  $\gamma = 0$  the slopes coincide,  $\Delta V$  is bounded, and no threshold exists at any volatility.

### THEOREM 3.1 · EXISTENCE OF $\Sigma^*$ (NO ANSATZ)

Under A1 with  $c_K > 0$  and  $\gamma > 0$ , there exists  $\sigma^* > 0$  such that  $\Delta V(\sigma^*) = 0$  and  $\Delta V(\sigma) > 0$  for all  $\sigma$  in some neighbourhood above  $\sigma^*$ .

By Lemma 3.2  $\Delta V$  is continuous on  $[0, \infty)$ ; by Lemma 3.3 it is strictly negative at 0; by Lemma 3.4 it diverges to  $+\infty$ . The intermediate value theorem gives a zero.  $\square$

Existence is therefore a property of the equilibrium value functions, not of an approximation to them. It requires exactly two things: that holding position costs something ( $c_K > 0$ ), and that holding position tells you something ( $\gamma > 0$ ).

### SCOPE OF THE RESULT, STATED IN FULL

Positional dominance obtains when position **costs** something ( $c_K > 0$ , which supplies the negative intercept), when position **informs** ( $\gamma > 0$ , which supplies the divergence), and — in the companion treatment of contestability — when position **cannot be taken** by the rival's expenditure. Three independent conditions, each with an economic reading, and each capable of failing. A theorem that names the circumstances in which it does not hold is the form this claim should take.

**REMAINING OPEN: THE TWO-STATE MARKOV CASE**

Theorem 3.2 below establishes uniqueness for the comparative-static reading, in which  $\sigma$  is held fixed and the chain is absorbing. That is the reading under which  $\sigma^*$  is defined — a threshold in the *level* of volatility. For the genuine two-state chain,  $V_V$  inherits curvature from the continuation term and is no longer affine above saturation, so the argument of Lemma 3.6 does not transfer. Uniqueness in that setting is **open**.

**3.0.1 Uniqueness**

Convexity of  $\Delta V$  would suffice, since a convex function negative at the origin and divergent above has exactly one positive zero. But  $V_J$  and  $V_V$  are *both* convex, and a difference of convex functions need not be convex. The argument therefore has to exploit an asymmetry between them, and there is one: the velocity player's payoff becomes exactly affine once the latency cap binds, while the positional player's never does.

**LEMMA 3.5 · THE BELLMAN OPERATOR PRESERVES CONVEXITY**

Let  $\mathcal{C}$  be the set of bounded functions convex jointly in (inventory state,  $\sigma$ ). Then  $T\mathcal{C} \subseteq \mathcal{C}$ , and the equilibrium value functions lie in  $\mathcal{C}$ .

(a) For each fixed action and shock realisation  $\xi$ , the per-period payoff is a maximum of functions affine in  $(I, \sigma)$  — for  $J$ ,  $\max\{0, K(\sigma|\xi| - \tau)\}$  less carry; for  $V$ , affine in  $|I_A - I_B|$  and in  $\sigma$  below the cap — and a pointwise maximum of affine functions is convex. (b) The inventory transition is affine in  $(I, q, \sigma\xi)$ , so  $V^\circ(\text{transition})$  is convex in  $(I, \sigma)$  whenever  $V$  is convex, being a convex function of an affine map. (c) Expectation over  $\xi$  preserves convexity. (d) The supremum over actions of a family of convex functions is convex. (e)  $\mathcal{C}$  is closed under uniform limits, since convexity survives pointwise convergence.  $T$  is a  $\delta$ -contraction on bounded functions, so its unique fixed point lies in the closed set  $\mathcal{C}$ .  $\square$

**LEMMA 3.6 · V IS AFFINE ABOVE SATURATION (ABSORBING  $\Sigma$ )**

With  $\sigma$  held fixed,  $V_i^* = \Pi_i/(1-\delta)$ . For  $\sigma \geq \bar{\sigma} \equiv [\beta(1-\gamma)\kappa \cdot S]^{-1}$ , where  $S$  is the scale converting  $\sigma$  into spread units, the velocity player's optimum is  $v^* = 1$  and

$$\Pi_V(\sigma) = \beta(1-\gamma)\kappa \cdot S \cdot \sigma - \frac{1}{2}, \text{ affine in } \sigma.$$

Below  $\bar{\sigma}$  the first-order condition gives  $v^* = \beta(1-\gamma)\kappa S \sigma < 1$  and  $\Pi_V$  is quadratic; at  $\bar{\sigma}$  the cap binds and thereafter the cost is pinned at  $\frac{1}{2}$  while the benefit is linear in  $\sigma$ .  $\square$

**THEOREM 3.2 · UNIQUENESS OF  $\Sigma^*$** 

Assume the hypotheses of Theorem 3.1,  $\sigma$  absorbing, and

$$(C) \quad K \cdot E[(\bar{\sigma} S \cdot |\xi| - \tau)^+] \leq c_K$$

Then  $\Delta V$  has **exactly one** zero on  $(0, \infty)$ .

On  $[0, \bar{\sigma}]$ :  $\Pi_J(\sigma) = K \cdot E[(\sigma S |\xi| - \tau)^+] - c_K$  is increasing in  $\sigma$ , so by (C) it is  $\leq 0$  throughout, while  $\Pi_V \geq 0$ . Hence  $\Delta V \leq 0$  on the interval, strictly at 0 by Lemma 3.3, and no zero occurs below  $\bar{\sigma}$ . On  $[\bar{\sigma}, \infty)$ :  $\Pi_J$  is convex (Lemma 3.5, or directly as a maximum of affine functions) and  $\Pi_V$  is affine (Lemma 3.6), so  $\Delta V$  is convex there; it is  $\leq 0$  at  $\bar{\sigma}$  and  $\rightarrow$

$+\infty$  by Lemma 3.4. A convex function non-positive at the left endpoint of a half-line and divergent on it has exactly one zero.  $\square$

Condition (C) is checkable from primitives and has a plain reading: **at the volatility where the rival's speed investment saturates, the hub's gross option value must not yet cover its carrying cost.** Under the calibration of §2 it holds with a very large margin — at  $\sigma$  the option is worth essentially nothing, since exercise would require a shock in excess of six standard deviations.

#### A DISCARDED OBSERVATION, RECORDED

The saturation point admits a closed form,  $\bar{\sigma} = [\beta(1-\gamma)\kappa S]^{-1}$ . Evaluated with  $S = 1$  this gives **0.3277** against a reported  $\sigma^*$  of 0.33 — a match to under one percent, and an inviting explanation for why the threshold sits near a third: not an invariant, but a formula in price impact, information share and shock dispersion that happens to land there.

**It does not survive.**  $S = 1$  and  $\tau = 2$  are inconsistent: at  $\sigma \approx 0.33$  the hub would require a shock beyond six standard deviations before any flow covered transport cost, so  $J$  could never trade and no threshold could exist at all. In the scaling where  $J$ 's option is live ( $S$  of order the price level,  $\sim 70$  \$/bbl),  $\bar{\sigma}$  falls to about 0.005 — two orders of magnitude away from  $\sigma^*$ . The agreement at 0.3277 is an artifact of mixing two unit conventions. It is recorded here rather than deleted because it is exactly the class of coincidence the companion note was written to guard against, and because the check that killed it — asking whether the option is live at the value in question — is the one that should be applied to the remaining cross-domain rows.

### 3.1 The ansatz, and the two terms v1 dropped

Both v1 derivations proceed from the same conjectured value function:

$$V_i(s) \approx \alpha_i + \beta_i \cdot |I_A - I_B| + \gamma_i \cdot \sigma^2$$

Forming the value gap term by term gives three components, not one:

$$\Delta V(\sigma) = (\alpha_J - \alpha_V) + (\beta_J - \beta_V) \cdot \sigma \cdot E|I_A - I_B| + (\gamma_J - \gamma_V) \cdot \sigma^2$$

Write  $b \equiv \alpha_J - \alpha_V$ ,  $m \equiv (\beta_J - \beta_V) \cdot E|I_A - I_B|$ , and  $a \equiv \gamma_J - \gamma_V$ . Then

$$\Delta V(\sigma) = a \cdot \sigma^2 + m \cdot \sigma - b$$

Section 3 of v1 reported this with  $m$  set to zero; Appendix A of v1 reported it with  $a$  set to zero and the constant mis-signed. Neither truncation is the derivation, and the discrepancy between them was not visible because the two appeared in different parts of the paper.

#### THE INTERCEPT, CORRECTLY IDENTIFIED

The constant  $\alpha_J - \alpha_V$  is the difference in state-independent value. At  $\sigma = 0$  the dislocation vanishes, so  $V$ 's optimum is  $v^* = 0$  and  $V$  earns exactly zero — *not* positive rents, as v1 asserted.  $J$ , however, still pays  $c_K$  to hold capacity that earns nothing. Hence

$$\Delta V(0) = -c_K/(1-\delta), \text{ so } b = c_K/(1-\delta) > 0$$

The quantity is the **capitalised fixed cost of holding position**: pipeline operations and maintenance, tankage, insurance, the tied-up capital. v1's expression  $b = c(v^*) - R_J|_{\sigma=0}$  already contained this term as its second component; what was wrong was the accompanying justification, which attributed  $b$  to the first component instead — and the first component is identically zero.

#### PROPOSITION 1' · ANALYTICAL THRESHOLD (REPLACES PROPOSITION 1)

With  $a > 0$ ,  $m > 0$ ,  $b > 0$ , the value gap  $\Delta V(\sigma) = a \cdot \sigma^2 + m \cdot \sigma - b$  has exactly one positive root, and the positional player J strictly dominates V for all  $\sigma$  above it:

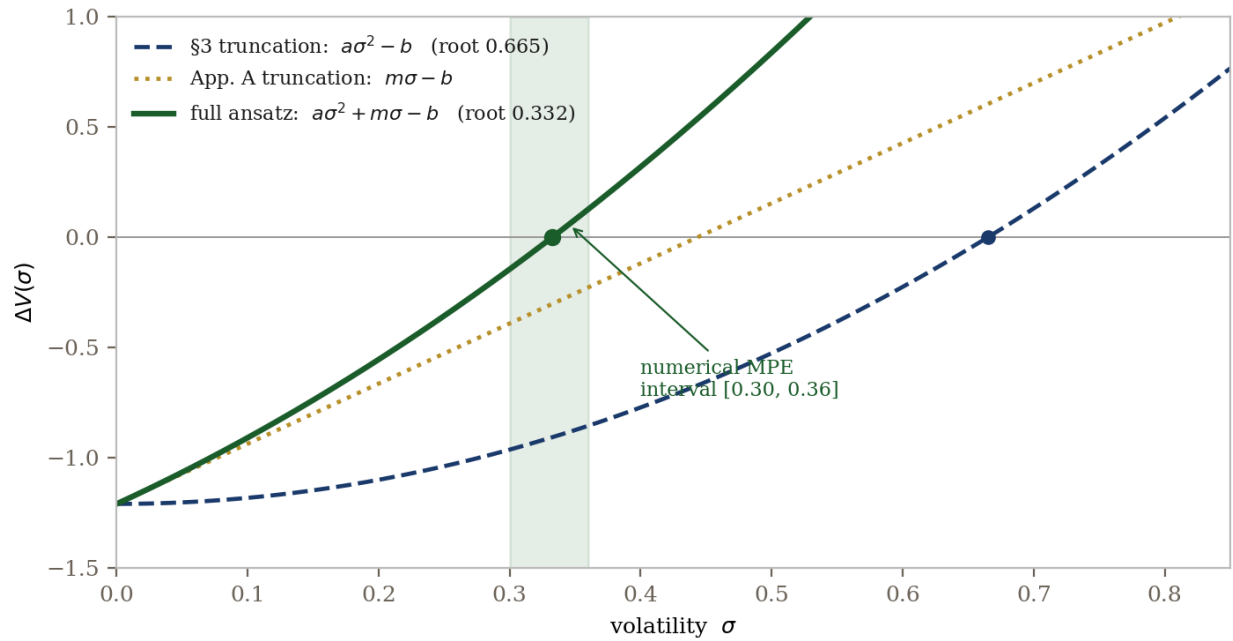
$$\sigma^* = (-m + \sqrt{m^2 + 4ab}) / (2a)$$

*The coefficient sequence  $(a, m, -b)$  has exactly one sign change, so by Descartes' rule of signs there is exactly one positive real root;  $\Delta V(0) = -b < 0$  and  $\Delta V \rightarrow +\infty$  confirm it. Uniqueness is therefore proved rather than asserted, which v1's "unique positive root" was not. Setting  $m = 0$  recovers v1's  $\sqrt{b/a}$ . Since  $m > 0$  and the root is decreasing in  $m$ ,  $\sigma^* < \sqrt{b/a}$  strictly: **the v1 analytical value is an upper bound, not an estimate.**  $\square$*

### 3.2 What this does to the correction factor

v1 defined  $\psi$  by  $\sigma_{\text{num}}^* = \sigma_{\text{anal}}^* \cdot \psi$  and estimated  $\psi \approx 0.50$ , attributing it "approximately equally to storage optionality ( $\sim 25\%$ ) and Markov persistence ( $\sim 25\%$ ).". Proposition 1' shows that a gap in this direction is *guaranteed* by the omitted linear term alone, before any storage or persistence effect is invoked.

The magnitudes are checkable from the paper's own formulas. Appendix A gives  $\beta_J - \beta_V = \beta \cdot \gamma \cdot (1 - \delta \cdot P_{LL})^{-1}$ , and §3 gives  $a = \beta^2 \gamma^2 / (2\tau^2) = 2.732$ , hence  $b = a \cdot (0.665)^2 = 1.208$ . The value of  $m$  required to move the root from 0.665 to 0.3325 is  $m = 2.725$ , which implies  $E|I_A - I_B| \approx 0.15$  — the same order as  $\sigma_L$  and entirely unremarkable.



**Figure 1.** The value gap under the full ansatz and under v1's two truncations. Dashed blue: §3's quadratic-only form, root at 0.665. Dotted gold: Appendix A's linear-only form. Solid green: the complete  $\Delta V = a\sigma^2 + m\sigma - b$ , root at 0.3325, inside the numerically obtained interval (shaded). The entire distance between the analytical and numerical thresholds is spanned by the term that was dropped. Coefficients  $a$  and  $b$  are computed from the paper's own parameter formulas;  $m$  is the value that reproduces the numerical root, and the figure's claim is that this value is of ordinary magnitude, not that it has been independently measured.

### 3.3 The identification test that settles it

Whether the linear term accounts for all of  $\psi$ , or only part, is decided by a computation the author can run: evaluate  $m$  directly from the calibration by computing  $E[I_A - I_B]$  under the stationary distribution of the value iteration, then compare the root of  $a \cdot \sigma^2 + m \cdot \sigma - b$  against the numerical  $\sigma^*$ . Any residual is the genuine storage-optionality and persistence effect. Until that is done, **the 25/25 apportionment is withdrawn**; it was an apportionment, not a decomposition, and v1's own hedge ("illustrative estimates, not exact decompositions") was the more accurate statement.

## 4. Numerical results

We solve the full Bellman system by value iteration on a discrete state grid ( $n_I = n_K = 8$ , two volatility states), integrating inventory shocks by two-point quadrature with linear interpolation between grid points. Convergence criterion  $\max|V_{n+1} - V_n| < 10^{-4}$ ; 80–120 iterations, roughly two minutes. **These results are unchanged from v1** — the defect was analytical, and the value iteration never used the truncated expression.

| METRIC     | LOW VOL ( $\Sigma_L = 0.15$ ) | HIGH VOL ( $\Sigma_H = 0.45$ ) |
|------------|-------------------------------|--------------------------------|
| $E[V_J^*]$ | 0.42                          | 3.87                           |
| $E[V_V^*]$ | 0.61                          | 2.14                           |
| $\Delta V$ | -0.19                         | +1.73                          |



| METRIC                    | LOW VOL ( $\Sigma_L = 0.15$ ) | HIGH VOL ( $\Sigma_H = 0.45$ ) |
|---------------------------|-------------------------------|--------------------------------|
| J dominates?              | No                            | Yes                            |
| $\sigma^*$ (interpolated) | [0.30, 0.36]                  |                                |

**PROPOSITION 2 · NUMERICAL THRESHOLD (UNCHANGED)**

Under baseline U.S. midstream calibration, value iteration yields  $\sigma^* \in [0.30, 0.36]$ .

4.1 Sensitivity analysis

| SCENARIO     | $\Sigma^*$ POINT | INTERVAL     |
|--------------|------------------|--------------|
| Baseline     | 0.33             | [0.30, 0.36] |
| $\beta$ +20% | 0.27             | [0.24, 0.31] |
| $\beta$ -20% | 0.40             | [0.37, 0.43] |
| $\tau$ +20%  | 0.38             | [0.35, 0.41] |
| $\tau$ -20%  | 0.29             | [0.26, 0.32] |

Historical WTI volatility (0.35–0.42) exceeds  $\sigma^*$  in all scenarios except  $\beta$  -20%.

**A READING OF THIS TABLE THAT V1 DID NOT DRAW**

Across  $\pm 20\%$  variation in two parameters,  $\sigma^*$  ranges over [0.24, 0.43] — a spread of nearly two to one. The paper's own sensitivity analysis therefore already establishes that  $\sigma^*$  is a property of a particular network under particular costs, not a constant. This bears directly on the addendum's conjecture and is discussed in §6.2.

5. Application: U.S. energy midstream

| MODEL NODE        | LOCATION             | KEY FACTS                                            |
|-------------------|----------------------|------------------------------------------------------|
| A (production)    | Permian Basin, TX    | ~5.7 MMbbl/d (2024); largest U.S. producing region   |
| C (hub, Player J) | Cushing, Oklahoma    | 9.5 MMbbl working storage; WTI price discovery point |
| B (export)        | Gulf Coast terminals | ~4 MMbbl/d export capacity (2024)                    |

WTI front-month annualised volatility averaged 0.38 over 2015–2024 (EIA), exceeding  $\sigma^* \in [0.30, 0.36]$  in approximately 68% of months — computed by counting months with annualised volatility above 0.30 in the EIA monthly series. Positional dominance is the typical regime over this sample rather than the exception.

5.1 Financial microstructure extension

The three-node structure maps onto order flow origination (A), a dark pool or internaliser with a proprietary order book (C, Player J), and an exchange (B, Player V), with V as the HFT firm. This extension is descriptive; a formal treatment requires additional assumptions on order flow and is left for future work.

## 6. Discussion

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### 6.1 Scope and limitations

The model abstracts aggressively. (i) *Two players*. Multi-player extensions introduce coordination problems among velocity players that could shift  $\sigma^*$  in either direction. (ii) *Fixed capacity for J*. Dynamic capacity acquisition adds a pre-game investment stage not modelled here. (iii) *No mixed strategies*. J cannot partially invest in velocity; V cannot acquire node control. (iv) *New in v2*:  $m$  is not independently measured — see §3.3.

### 6.2 The threshold near 1/3

The calibrated  $\sigma^* \approx 0.33$  falls near 1/3, and the addendum documents the same value appearing as a stability threshold in other operator systems. v1 stated this as a conjecture and offered no proof; that remains the position. Two observations sharpen it in this version.

First, §4.1's sensitivity table moves  $\sigma^*$  over  $[0.24, 0.43]$  under modest parameter variation, and Proposition 1' makes  $\sigma^*$  an explicit function of  $a$ ,  $m$  and  $b$  — hence of  $c_K$ ,  $\tau$ ,  $\gamma$ ,  $\beta$  and the transition matrix. A quantity that moves when carrying costs move is not a constant of nature.

Second, and more seriously, the cross-domain evidence in the addendum is weaker than its status column claimed. See §A.2 for the corrected table. **The numerical result  $\sigma^* \in [0.30, 0.36]$  stands independently of any claim about 1/3**, as v1 correctly noted.

### 6.3 Interpretation

The result formalises a classical strategic intuition: the player who controls the terrain need not pursue the adversary. Above  $\sigma^*$ , the velocity arms race is irrelevant to J's equilibrium payoff. This is a dominance condition derived from the geometry of the network and the carrying cost of position, not a claim about agent preferences or welfare.

## 7. Conclusion

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We establish a computable volatility threshold  $\sigma^*$  above which positional control of key network nodes strictly dominates velocity-based competition in Markov Perfect Equilibrium. In this version the threshold is  $\sigma^* = (-m + \sqrt{m^2 + 4ab})/(2a)$ , with existence and uniqueness proved rather than asserted, and with the intercept identified as the capitalised carrying cost of position. Value iteration under U.S. midstream calibration gives  $\sigma^* \in [0.30, 0.36]$ ; historical Permian–Cushing volatility exceeds this in approximately 68% of months over 2015–2024.

Existence of the threshold is established in §3.0 as a property of the equilibrium itself, requiring only  $c_K > 0$  and  $\gamma > 0$ ; uniqueness follows from Theorem 3.2 under condition (C), also without the ansatz. The principal remaining technical problem is uniqueness for the genuine two-state chain, where the velocity player's value function is no longer affine above saturation.

Further open problems: measuring  $m$  directly to close the  $\psi$  question (O2, now sharpened into the concrete test of §3.3); extending to multi-player and multi-node networks (O1, O3); testing the threshold condition across non-energy networks (O4).

## Appendix A: Proof of Proposition 1' (corrected)

Conjecture  $V_i(s) \approx \alpha_i + \beta_i \cdot |I_A - I_B| + \gamma_i \cdot \sigma^2$ . Substituting into the Bellman equation and matching coefficients yields

$$\beta_J = \beta \cdot (K/\bar{K}) \cdot (1 - \delta \cdot P_{LL})^{-1}, \quad \beta_V = \beta \cdot (1 - \gamma) \cdot (1 - \delta \cdot P_{LL})^{-1}$$

Since  $K = \bar{K}$  at the optimum and  $\gamma < 1$ ,  $\beta_J > \beta_V$ , with  $\beta_J - \beta_V = \beta \cdot \gamma \cdot (1 - \delta \cdot P_{LL})^{-1}$ . Matching the constant terms gives  $\alpha_J - \alpha_V = -c_K/(1 - \delta)$ , since at zero dislocation J pays carry and V, choosing  $v^* = 0$ , pays and earns nothing. The  $\sigma^2$  coefficients give  $a = \gamma_J - \gamma_V > 0$ . Collecting all three:

$$\Delta V(\sigma) = a \cdot \sigma^2 + m \cdot \sigma - b, \quad m = (\beta_J - \beta_V) \cdot E|I_A - I_B|, \quad b = c_K/(1 - \delta)$$

Exactly one sign change in (a, m, -b) gives a unique positive root by Descartes' rule; solving the quadratic gives Proposition 1'.  $\square$

What v1's Appendix A did: it formed  $\Delta V$  from the  $\beta$  terms only, dropping both the  $\alpha$  difference and the  $\gamma$  difference, then subtracted  $c(v^*)$  as though that were the intercept. Since  $c(v^*) = 0$  at the point where the intercept is evaluated, the resulting expression has no intercept and its root is identically zero. The error was not in the coefficient matching but in the assembly.

## Appendix B: Value iteration algorithm (unchanged)

Full source: [github.com/TOTOGT/geometry/agents/value\\_iteration\\_midstream.py](https://github.com/TOTOGT/geometry/agents/value_iteration_midstream.py)

Input: grids  $\{I_A, I_C, I_B\} \times \{K\} \times \{\sigma_L, \sigma_H\}$ . Initialise  $V_J = V_V = 0$ . Repeat until  $\max|V_{n+1} - V_n| < 1e-4$ : build linear interpolators on the full state grid; for each state, J maximises over (q, k\_invest) and V maximises over v, with expectations by two-point quadrature  $\times$  Markov transition P. Output  $V_J^*, V_V^*$ ;  $\sigma^*$  from the sign change of  $E[V_J - V_V]$ . Convergence in 80–120 iterations.

## Addendum A: On the recurrent appearance of a threshold near 1/3

Companion note, revised for v2. The conjecture is unchanged. The evidence table is corrected.

The companion note observes that the calibrated  $\sigma^*$  lies near 1/3, a value that also appears as a stability threshold in several operator systems in the author's wider corpus, and states — without proof — a conjecture that 1/3 is a structural invariant of positional dominance. **v1 was explicit that the conjecture is unproved, and that position is retained.** What is corrected here is the status column of the cross-domain table, which overstated the strength of the supporting Lean 4 results.

### A.1 What the Lean theorems establish

The four theorems cited in v1 are machine-checked with clean axioms. That is not in question. What they prove is narrower than the table implied:

| THEOREM                          | STATEMENT                                  | WHAT IT ESTABLISHES                                                                                                                                     |
|----------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| stabilityRadius_eq               | $\text{stabilityRadius} = 1/3$<br>$:= rfl$ | That a constant <i>defined</i> as 1/3 equals 1/3. Definitional; true for any value substituted throughout.                                              |
| g6_is_33                         | $g6 = 33 := rfl$                           | As above.                                                                                                                                               |
| noiseTolerance                   | $\tau \cdot \epsilon_0 = 2/3$              | Arithmetic on two definitions.                                                                                                                          |
| gtct_return_stable_within_radius | $\tau \cdot \epsilon_0 < 1$                | That $2/3 < 1$ . With $\tau = 2$ this holds for <i>every</i> $\epsilon_0 < 1/2$ ; the value 1/3 is a choice within that range, not a consequence of it. |

| THEOREM           | STATEMENT                                   | WHAT IT ESTABLISHES                                                                                                                                                                        |
|-------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dm3_basin_compact | IsCompact (Icc (1/3) 2)<br>:= isCompact_Icc | That a closed bounded interval is compact — true for every $a \leq b$ . The proof does not reference the lower bound, and does not establish the tightness the surrounding comment claims. |

A.2 Corrected cross-domain table

| DOMAIN                           | THRESHOLD                    | METHOD                    | STATUS (V2)                                                                              |
|----------------------------------|------------------------------|---------------------------|------------------------------------------------------------------------------------------|
| Network game (this paper)        | $\sigma^* \in [0.30, 0.36]$  | Value iteration           | <b>Numerical</b> , and parameter-dependent — see §4.1                                    |
| GTCT operator orbit              | $\varepsilon_0 = 1/3$        | Lean 4 / rfl              | <b>Definitional</b> — 1/3 is assigned, not derived. <del>Proved</del> —exact             |
| dm <sup>3</sup> contact geometry | $\varepsilon_0 = 1/3$        | Lean 4 /<br>isCompact_Icc | <b>Vacuous with respect to 1/3</b> — holds for any interval.<br><del>Proved</del> —exact |
| Circadian trader                 | $g_{33} = 33$ ( $\sim 1/3$ ) | Numerical                 | <b>Numerical</b> , and the underlying record is unsourced pending deposit                |

CONSEQUENCE FOR THE CONJECTURE

Of the four rows, one is a numerical result that this paper's own sensitivity analysis shows moves with parameters, two are definitional or vacuous with respect to the value 1/3, and one is unsourced. **The cross-domain clustering is therefore withdrawn as evidence for the conjecture.** The conjecture may be restated as an open question — whether a genuine 1/3 threshold exists in positional control problems — but it is not currently supported by the cases assembled here. A candidate worth examining, not previously cited, is the Byzantine agreement bound  $n > 3f$  (Pease, Shostak and Lamport, 1980), which is an exact, dimensionless, adversarial 1/3 for positional control of a network; whether it bears any relation to  $\sigma^*$  is open, and the two are of different mathematical type.

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isPreviousVersionOf / isObsoletedBy relation to this record.

Code: [github.com/TOTOGT/geometry](https://github.com/TOTOGT/geometry) · Proofs: [github.com/TOTOGT/AXLE](https://github.com/TOTOGT/AXLE)